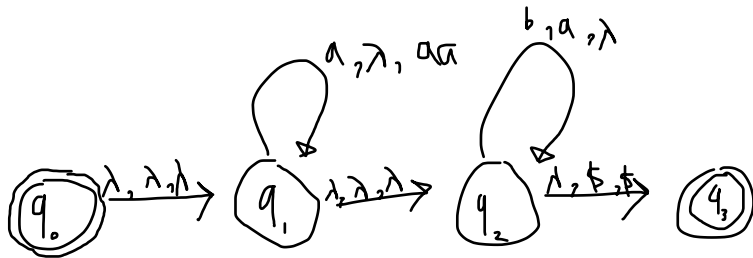


Theory of Computation

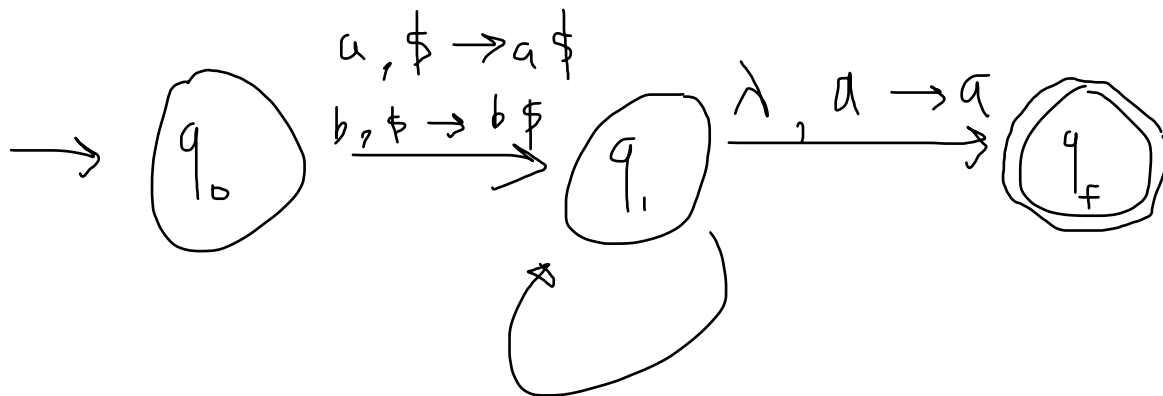
Exercise 10: (Pushdown Automata)

Find Pushdown Automata for the following languages.

1. $L_1 = \{ a^n b^{2n} : n \geq 0 \}$



$$2. L2 = \{w \in \{a, b\}^* : n_a(w) > n_b(w)\}$$



$$a, a \rightarrow aa$$

$$a, b \rightarrow \lambda$$

$$b, a \rightarrow \lambda$$

$$b, b \rightarrow bb$$

$$a, \$ \rightarrow a \$$$

$$b, \$ \rightarrow b \$$$

*3. $L3 = \{a^n b^m a^{n+m} : n, m \geq 1\}$

(Homework 8)

